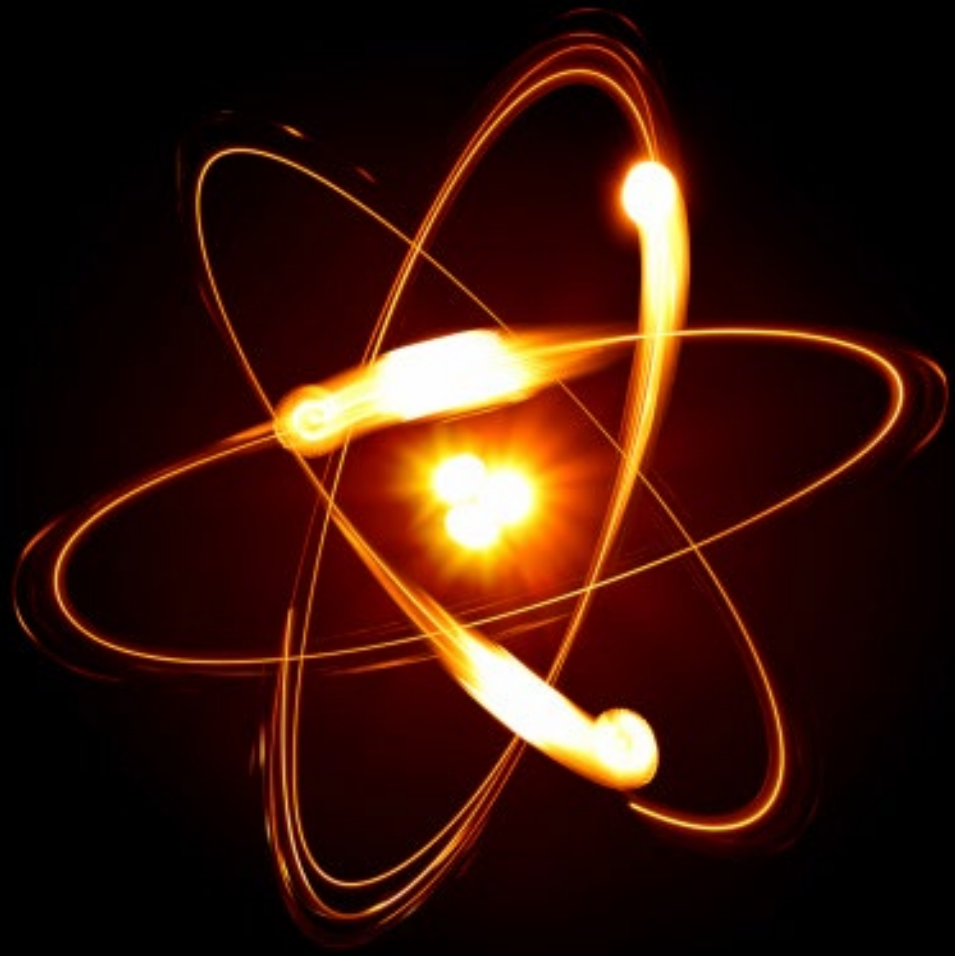
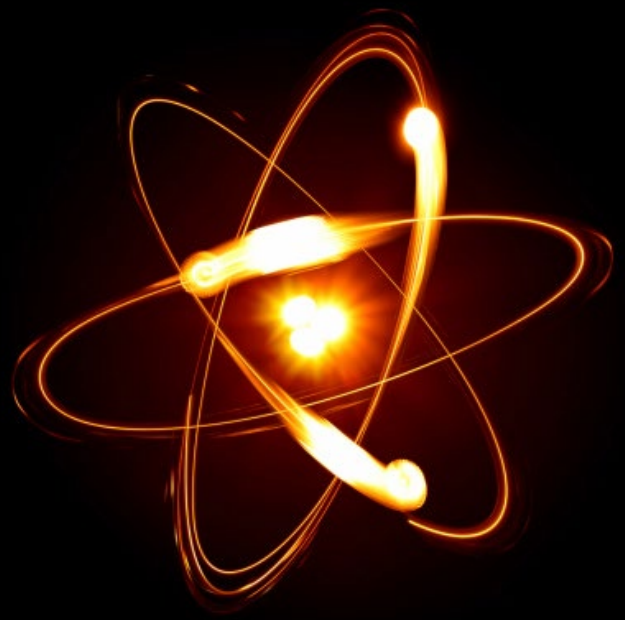
Chapter 3.

de Broglie´s postulate. Wave-like properties of particles



Beatriz González del Río

Quantum Physics



Chapter 3.

de Broglie´s postulate. Wave-like properties of particles

- The wave-particle duality
- Young´s two-slit interference experiment
- Taylor´s experiment
- Wave-particle duality. Einstein´s interpretation
- Wave-particle duality. De Broglie´s hypothesis & Bragg relation
- Davison and Germer´s experiment
- Heisenberg´s uncertainty principle
- Matter waves

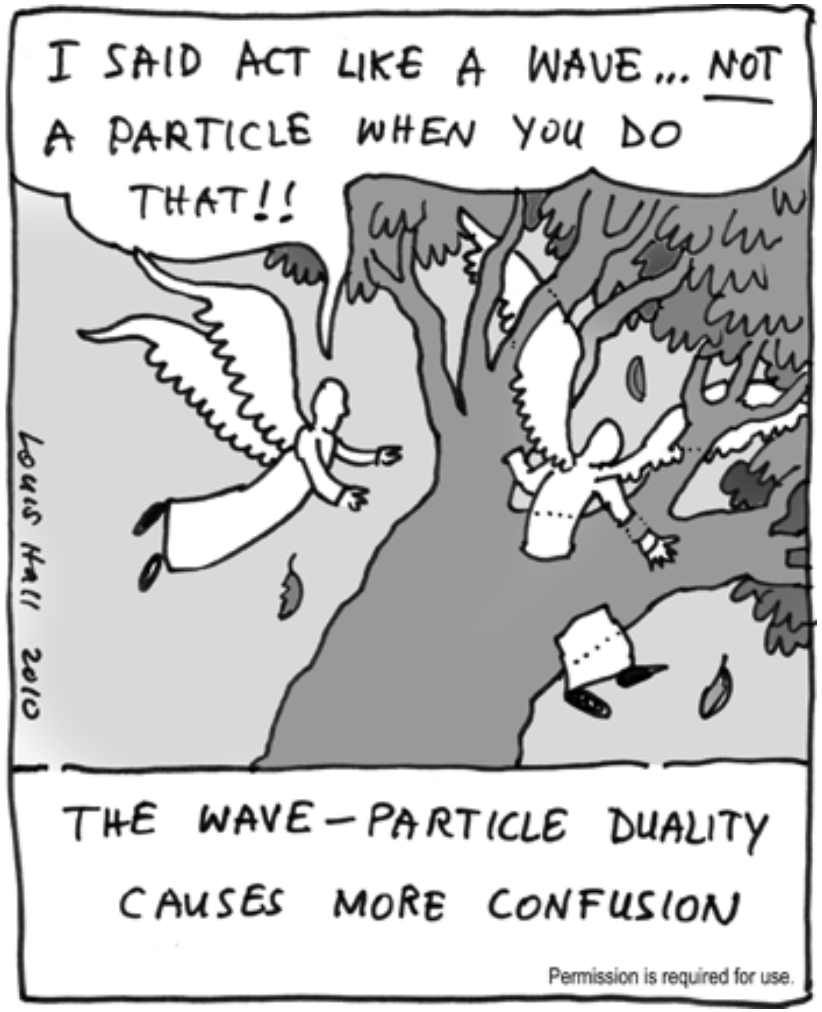
The Wave-Particle Duality

The previous chapter showed some experiments whose explanation required the electromagnetic radiation to be considered as made of some particle-like objects called photons.

On the other hand, the explanation of the interference and diffraction patterns requires to use a wave-like model for the electromagnetic radiation (Maxwell's equations)

Which model is the REAL one?

Both models are complementary. A complete interpretation of all phenomena can be obtained only by conserving **BOTH** the wave aspect and the particle aspect of radiation.



Young's two-slit interference experiment

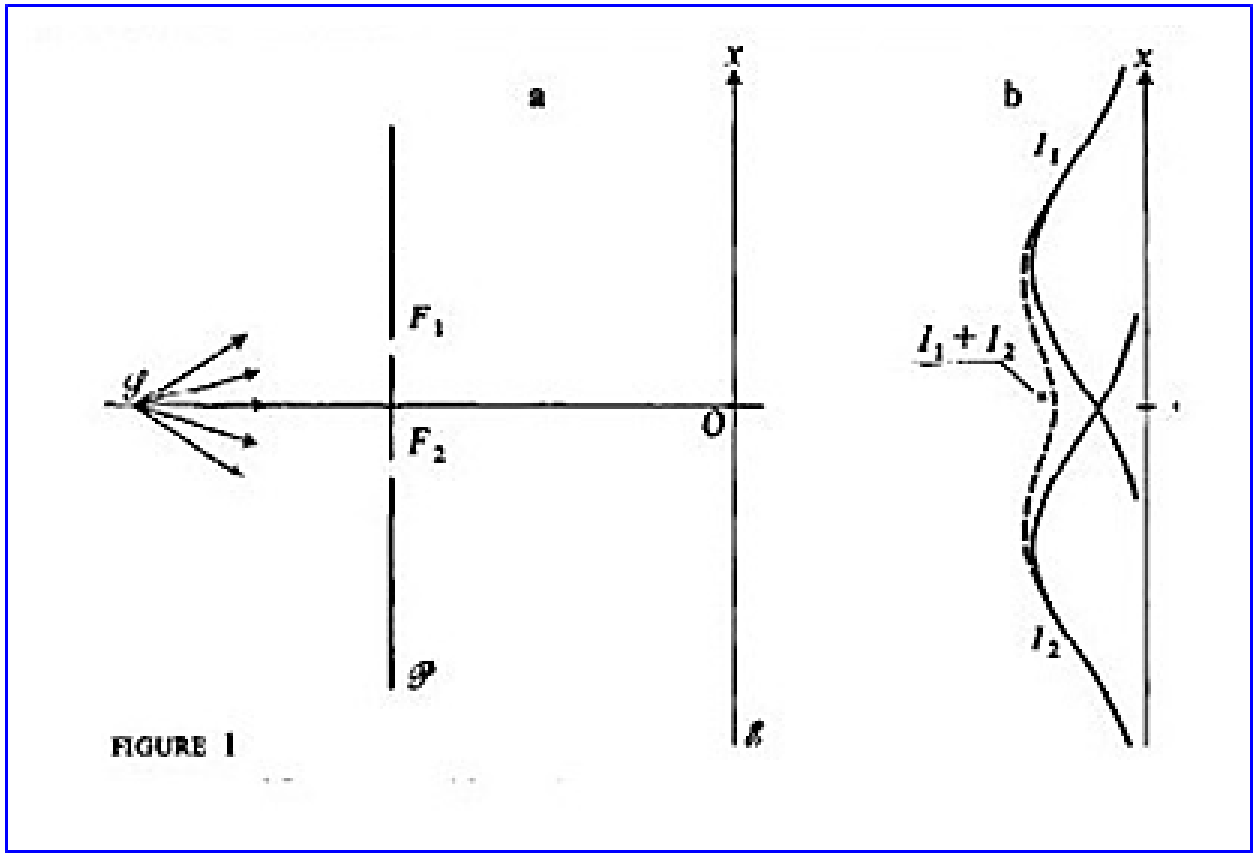


FIGURE 1

Monocromatic light falls on an opaque screen which has two narrow slits, F_1 and F_2, which illuminate the observation screen $\mathcal{E}$

If we block F_1 we obtain $I_2(x)$

If we block F_2 we obtain $I_1(x)$

If both are open?

$I_1(x) + I_2(x)$?

Young's two-slit interference experiment



Courtesy Jearl Walker

Figure 35-9 A photograph of the interference pattern produced by the arrangement shown in Fig. 35-8, but with short slits. (The

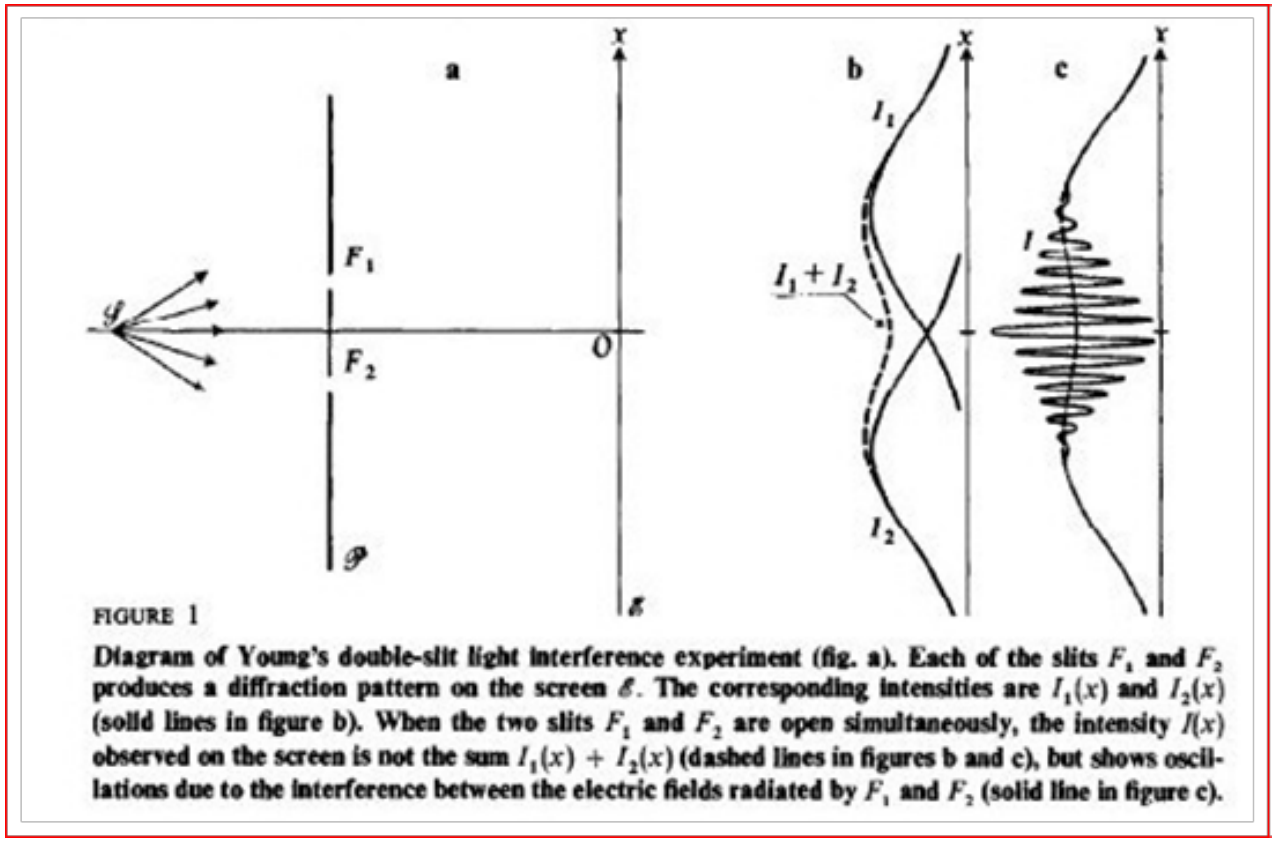


FIGURE 1

Diagram of Young's double-slit light interference experiment (fig. a). Each of the slits F_1 and F_2 produces a diffraction pattern on the screen $\mathcal{E}$. The corresponding intensities are $I_1(x)$ and $I_2(x)$ (solid lines in figure b). When the two slits F_1 and F_2 are open simultaneously, the intensity $I(x)$ observed on the screen is not the sum $I_1(x) + I_2(x)$ (dashed lines in figures b and c), but shows oscillations due to the interference between the electric fields radiated by F_1 and F_2 (solid line in figure c).

If both are open? $I(x) \neq I_1(x) + I_2(x)$

Young's two-slit interference experiment. Classical explanation

According to the wave-like model.....

The result can be explained within the classical (wave-like) model:

$$I_1(x) \propto |E_1(x)|^2$$

$$I_2(x) \propto |E_2(x)|^2$$

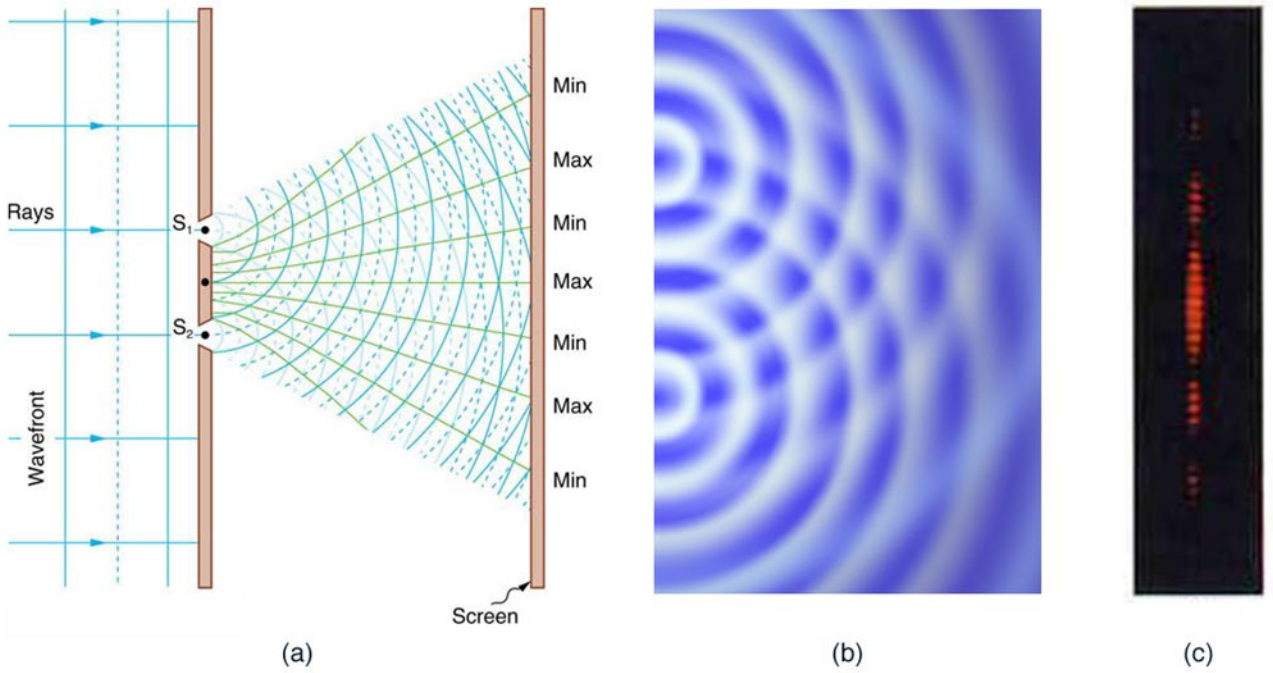
where $E_1(x)$ y $E_2(x)$ are the amplitudes of the electric field created at point x by the slits 1 and 2 respectively.

Now,

$$I(x) \propto |E_1(x) + E_2(x)|^2$$

and therefore

$$I(x) \neq I_1(x) + I_2(x)$$



What happens if the intensity is diminished?

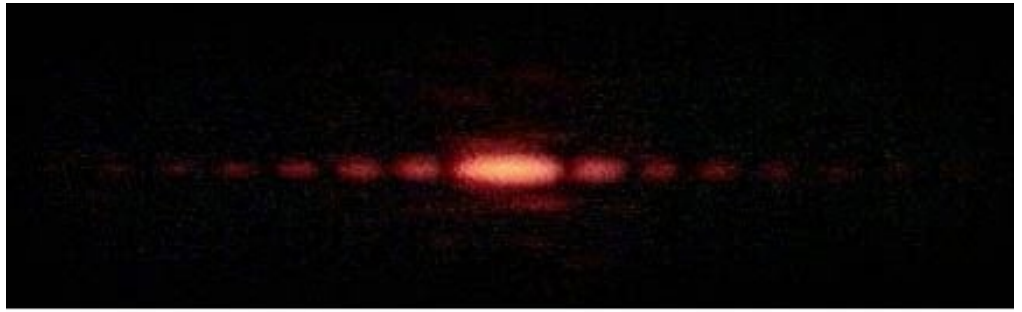
The interference pattern remains, although very weak.

Young's two-slit interference experiment. Corpuscle explanation

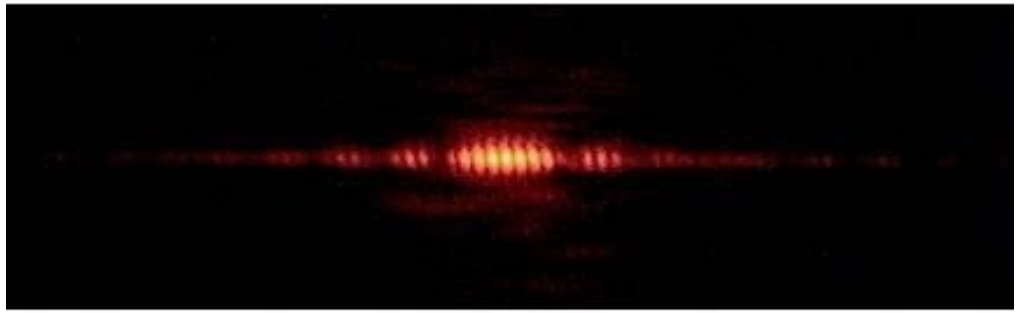
According to the particle-like model.....

The existence of a diffraction pattern (when only one of the slits is open) could be explained as due to *collisions* of the photons with the edges of the slit.

The existence of an interference pattern could be explained in terms of an interaction among the photons passing through slit 1 and those passing through slit 2.



Laser light passing through a single slit, showing basic diffraction pattern.



Double-slit pattern. Note the interference fringes, on top of the diffraction pattern.

What happens if the intensity is diminished, namely that the photons pass one by one?

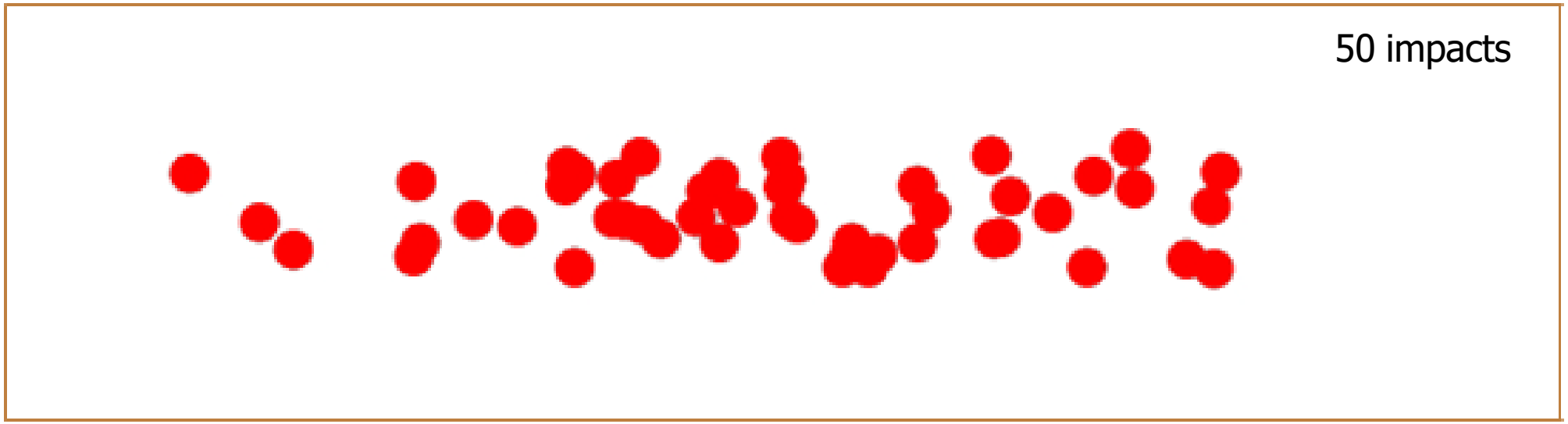
The interaction among them should disappear and therefore the interference pattern should disappear.

Taylor´s experiment

In 1909 Taylor carried out the experiment. He used an extremely weak light source. The photographic plate had to be exposed for several months. The results showed that:

- If the photographic plate **is exposed** for a **very short** time, so that it receives a few photons, we observe **localized impacts, but not a weak interference pattern**.

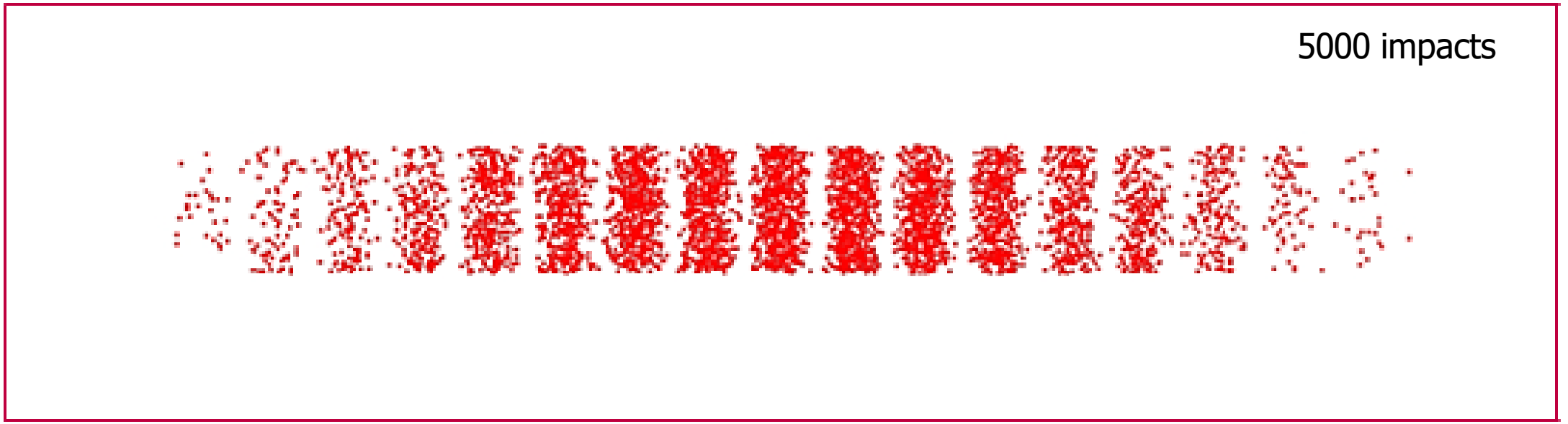
We must discard a purely wave interpretation of radiation.



Taylor´s experiment

- If the photographic plate **is exposed** for a time **long enough** so that it can receive many photons, **the interference pattern is clearly visible**.

Discards a purely corpuscle interpretation of radiation.

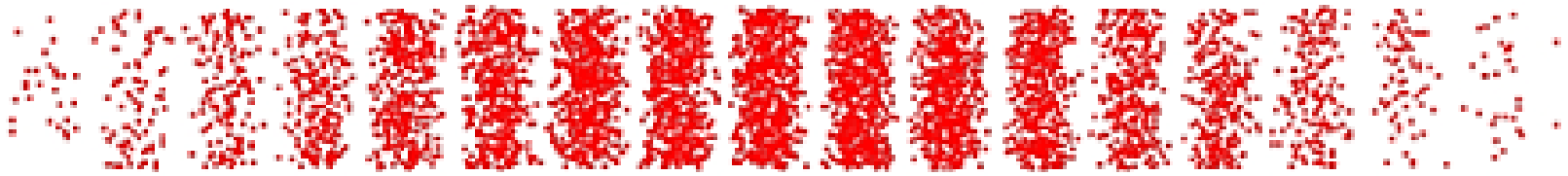


Taylor´s experiment

50 impacts



5000 impacts



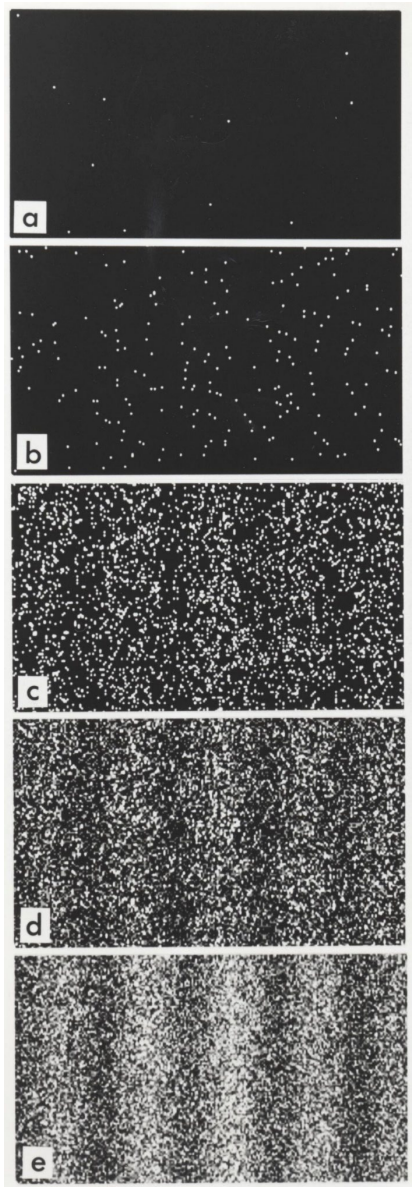
Taylor´s experiment

Summary

- **If exposed for very short time ⟶ localized impacts, not weak interference pattern ⟶ Must discard a purely wave interpretation of radiation.**

- **If exposed for long enough ⟶ interference pattern clearly visible Discards a purely corpuscle interpretation of radiation.**

- The individual impacts ⟶ **random**
- Great number of photons ⟶ distribution of impacts acquire a **continuous** aspect.



Taylor´s experiment. Other curiosities

- The density of impacts at a point x is proportional to $I(x)$. The photons build up the interference pattern as more and more arrive to the plate.

- If one slit is blocked, the photons build up the diffraction pattern. If the photons arrive one by one, and we consider a photon passing through one of the slits, why should the fact that the other is open or closed have such a critical importance?

How can we determine through which slit each photon passes?

PERFORM AN EXPERIMENT

But:

If we try to determine through which slit each photon passes, then we destroy the interference pattern ⇒ if we perform a measurement on the system, **we perturb the system in a fundamental way** ⇒
the measurement disturbs the system we pretend to study!!!!.

Given that it is not possible to observe the interference pattern and SIMULTANEOUSLY know the slit through which the photon passes then **it does NOT make any sense to say that the photon passes through a slit** ⇒
Does it make sense to talk about a trajectory?

Taylor´s experiment. Other curiosities

When the photons arrive one by one, their impacts on the screen produce gradually the interference pattern.

So, for a given photon, we do not know in advance where it will strike

But all photons are emitted under the same conditions !

CLASSICAL IDEA: The initial conditions completely determine the subsequent motion.....

IT DOES NO LONGER HOLD !!!!!!!

When a photon is emitted, we can ONLY say that the probability of it striking the screen at x is proportional to the intensity $I(x)$, calculated using the wave theory, i.e., to $|E(x)|^2$.

Wave-particle duality

To describe the peculiarity of photons of exhibiting properties of both waves and particles, physicists have introduced the ....

Wave-particle duality concept

- It is impossible to observe **simultaneously** the wave and particle aspects of light (electromagnetic radiation)

- The particle and wave aspects of light are inseparable. We need both aspects in order to explain **all the results**. Light behaves simultaneously like a wave and like a flux of particles, the wave enabling us to calculate the probability of the manifestation of a particle.

- Predictions about the behavior of a photon can only be probabilistic.

- When light **travels**, it behaves like a **wave** producing the interference and diffraction patterns. But when light **is detected** (interacts with matter) then it behaves like a localized **corpuscle**.

- The information about a photon at time t is given by the wave $E(r, t)$ which is a solution of Maxwell's eqns. $E(r, t)$ is interpreted as the probability amplitude of a photon appearing at time t at the point r. This means that the corresponding probability is proportional to $| E(r, t) |^2$.

Wave-particle duality. Einstein´s interpretation

It was Einstein who first proposed a statistical interpretation of the wave-particle duality for radiation.

The photons are produced in identical conditions. But they do not behave classically in the sense that the initial conditions DETERMINE the future behavior. This is so because their impact on the plate is random (the photons have a statistical behavior).

On the other hand, according to the wave model the intensity of the radiation at point $\vec{r}$, is proportional to $|E(\vec{r})|^2$.

Einstein's suggested that $|E(\vec{r})|^2$ is proportional to the average number of photons per unit volume at point $\vec{r}$.

Or, in other words, $|E(\vec{r})|^2$ is proportional to the **probability** that a photon be detected at point $\vec{r}$.

Wave-particle duality. Einstein´s interpretation

Consider a point source F.

According to the wave model, we have spherical waves whose intensity goes as $1/r^2$.

According to Einstein, the source emits photons randomly in all directions.

The area of the sphere centered at F and with radius r increases proportional to r^2. So, the average number of photons crossing a unit area will decrease (as r^2) with increasing distance from the source F.

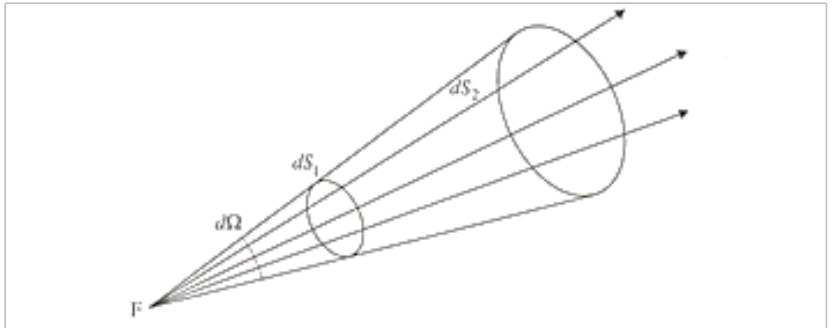


Figura 3.2. El número promedio de fotones que atraviesan las áreas dS_1 y dS_2 correspondientes a un ángulo sólido $d\Omega$ es el mismo, independientemente de su distancia a la fuente *F*. Obsérvese, no obstante, que el número de fotones por *unidad de área* que cruzan dS_1 y dS_2 es inversamente proporcional al valor de dichas áreas.

This agrees with the classical idea that the intensity of the radiation decreases as r^2 which increasing distance. Assuming that each photon carries an energy $h\nu$, then we can write

$$I(r) = \epsilon_0 |E(r)|^2, \quad (\textit{wave - like picture})$$

$$I(r) = N(r)h\nu, \quad (\textit{particle - like picture})$$

where $N(r)$ is the average number of photons per unit area and unit time and ϵ_0 is the dielectric constant of vacuum.

Identifying both expressions

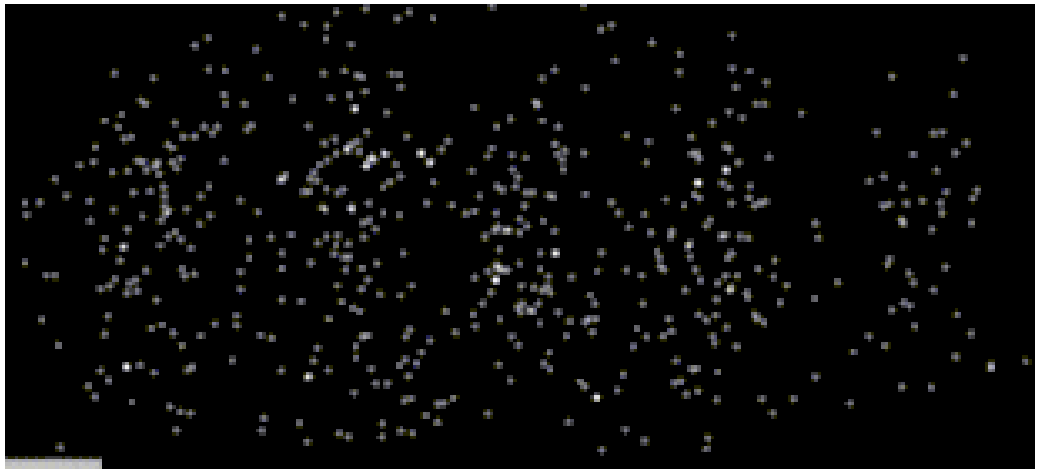
$$|E(r)|^2 \propto N(r)$$

which agrees with Einstein's interpretation and establishes the connection between the wave-like and particle-like models of radiation.

Wave-particle duality. Einstein´s interpretation

The waves associated to the classical model can be interpreted as some kind of guiding-waves for the photons, and the intensity of the wave in any given point gives an estimate of the average number of photons at that point.

When a photon reaches the plate after crossing the double-slit, it "lands" in some places **with greater probability than in other places, according to the respective values of** $I(x)$.



Wave-particle duality. de Broglie´s hypothesis

Louis de Broglie (1924) suggested a symmetrical interpretation for material particles. He proposed that material particles, just like photons, have a wave-like aspect. He associates a matter wave to the material particle



$$\lambda = \frac{h}{p}$$

λ is the de Broglie wavelength

Moreover, he also suggests

$$E = h\nu$$

These formulas are valid for the relativistic and non-relativistic cases (different relation between E and p).

Wave-particle duality. de Broglie´s hypothesis

The previous eqns. suggest a complete symmetry between the particle-like and wave-like models.

The similarity between the photon and a particle with mass is not complete: For the photon $E = pc$

If the particle has a mass *m* and moves with a velocity $v \ll c$ $E = \frac{p^2}{2m}$

In the relativistic case there is a somewhat different relationship.

What are the typical λ values in the macroscopic world? A dust particle of 10^{-2} g moving with v = 10 m/s.

$$\lambda = \frac{h}{p} = \frac{6.626\ 10^{-34}\ J\ s}{10^{-5}\ Kg\ \ 10\ m/s} = 6.626\ 10^{-30}\ m = 6.626\ 10^{-20}\ \text{Å}$$

IMPOSSIBLE TO DETECT

Consider a neutron moving with the same velocity

$$\lambda = \frac{6.626\ 10^{-34}\ J\ s}{1.68\ 10^{-27}\ Kg\ \ 10\ m/s} = 3.92\ 10^{-8}\ m = 3.92\ 10^{2}\ \text{Å}$$

CAN BE DETECTED

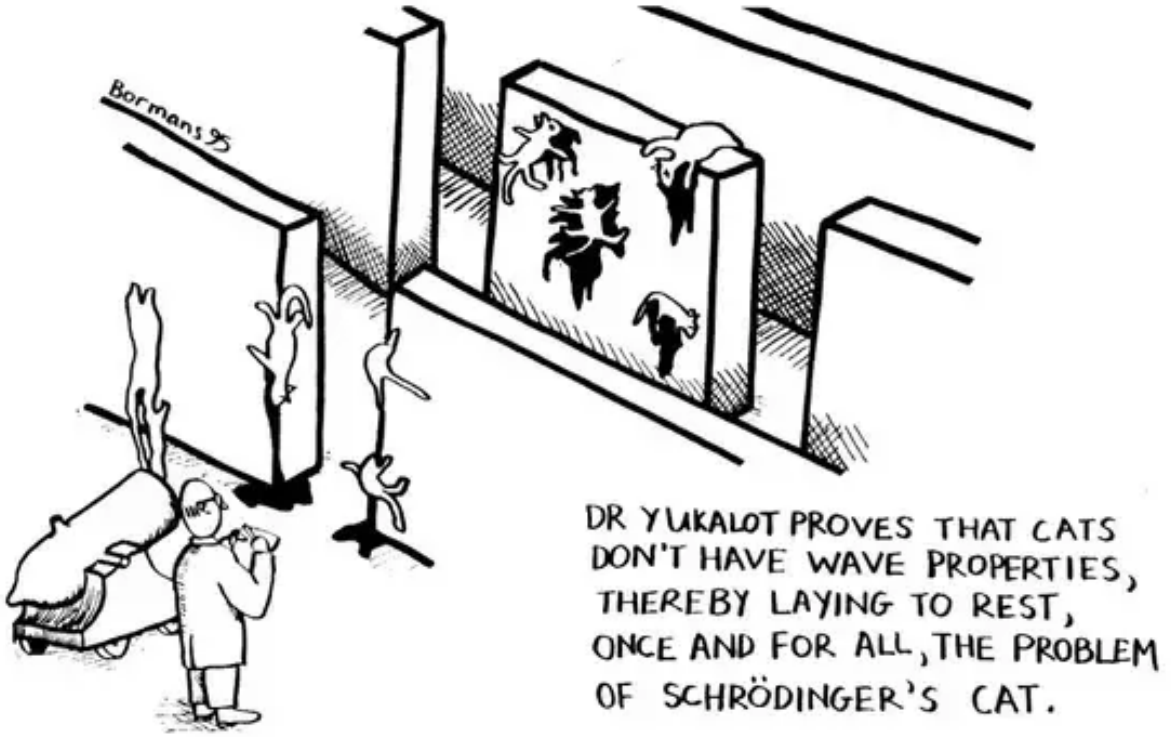
Determine the wavelength of an electron which has been accelerated by a voltage *V* = 100 volts.

Wave-particle duality. de Broglie´s hypothesis

If the moving particles have an associated wave, then it should be possible to produce the interference and diffraction patterns.

This is achieved when the obstacles or slits have dimensions comparable to the wavelength.

Therefore, in case of macroscopic objects, it is impossible to produce those patterns.



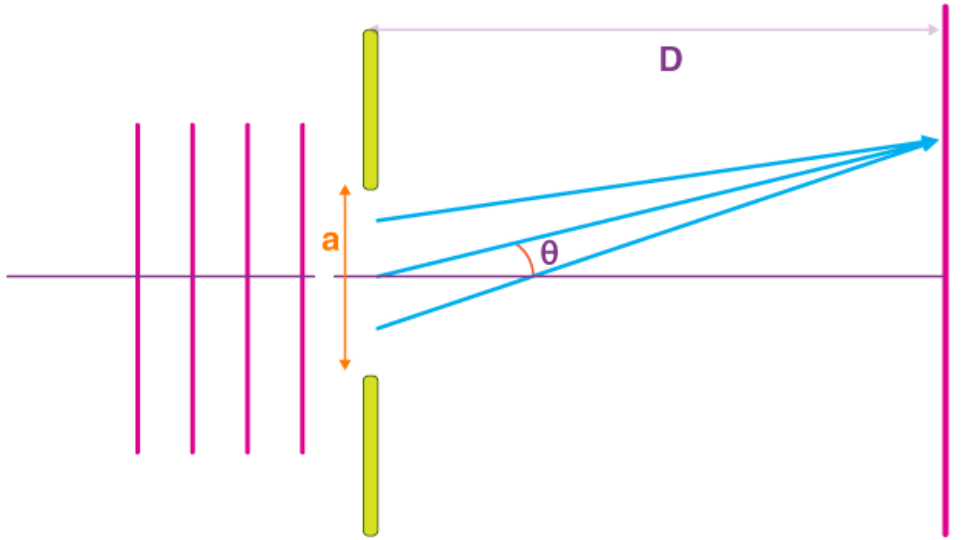
For microscopic particles, it is possible to produce those patterns.

Wave-particle duality in material particles

The wave nature of light (electromagnetic radiation) is NOT revealed by experiments in geometrical optics BECAUSE the dimensions of the instruments (characterized by a length *a*) are very large compared with λ.

$\lambda/a \rightarrow 0$ (*geometric optics*)

Diffraction effects are confined to angles $\theta \approx \lambda/a$



To observe wave-like aspects in the motion of matter we need systems with slits or obstacles of suitable small dimensions.

The smaller available systems are the crystalline solids where the spacing between adjacent atoms is such that a ≈ few Å

Elsasser suggested in 1926 that the wave-like nature of matter might be tested by sending a beam of electrons against a crystalline solid. We should observe similar images as those obtained with X-rays.

Experiments: Davisson and Germer (USA), Thomson (UK)

Wave-particle duality in material particles. Bragg relation

In a crystalline solid (ordered), the atoms can formally be arranged in groups of atomic planes with a separation distance *d*.

When light (radiation) falls on them, the atoms scatter the radiation.

The net effect is the same as if the radiation were reflected by the successive atomic planes. The total reflected light will be the superposition (interference) of those parts reflected by the different atomic planes.

Constructive interference (Diffraction maximum):

$$n\lambda = 2d \sin\theta$$

with *n* an integer number and λ the incident wavelength

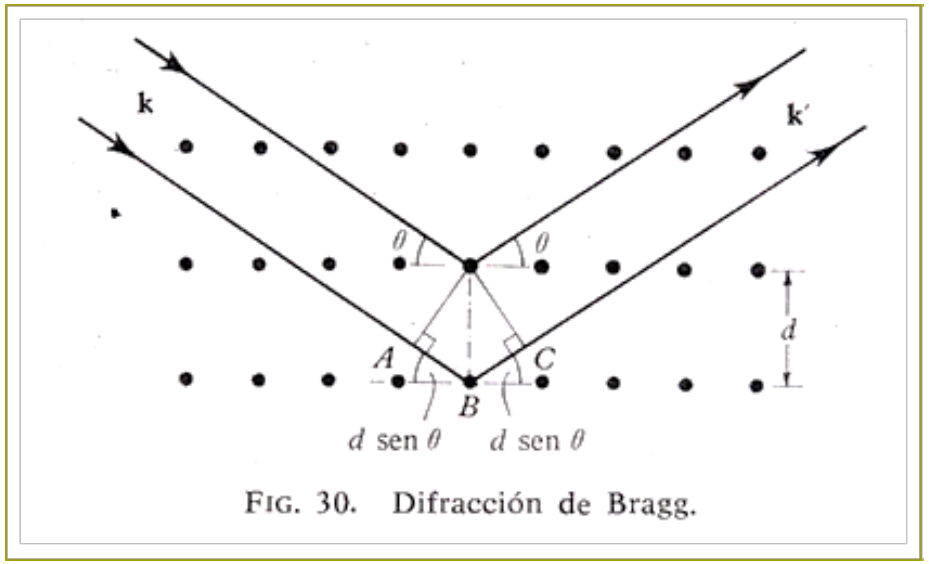
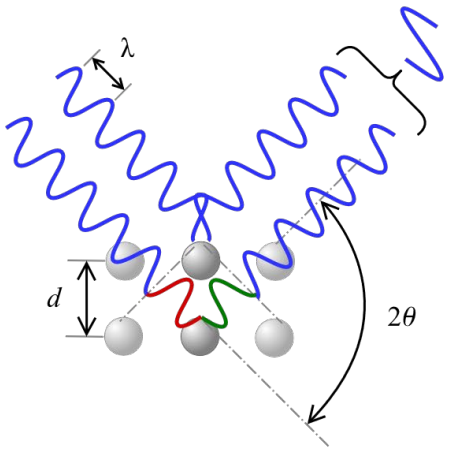
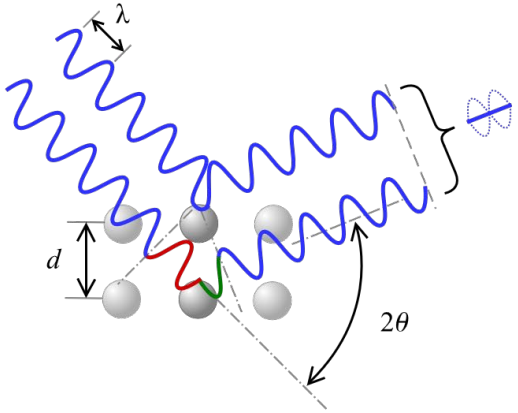


FIG. 30. Difracción de Bragg.





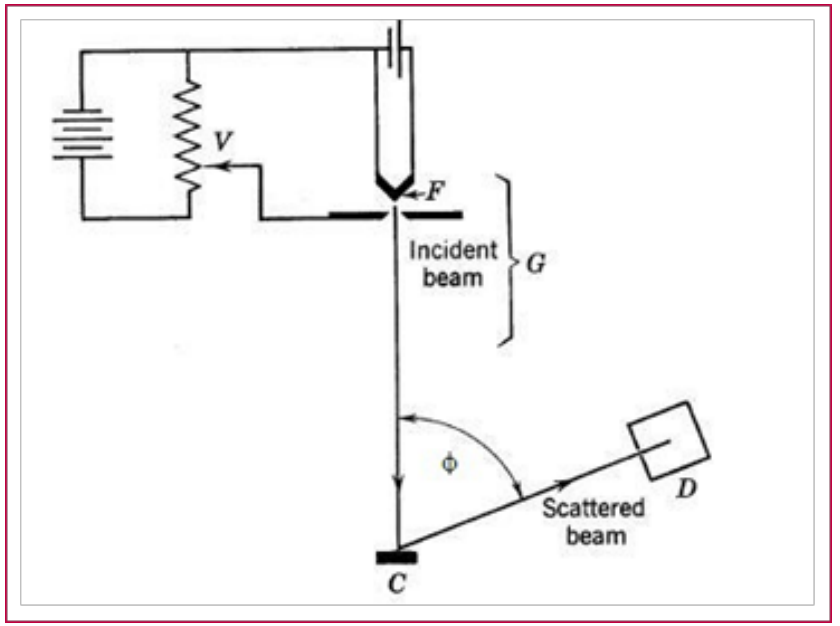
Davisson and Germer's experiment (1927)

Electrons are emitted by filament F and are accelerated by a potential difference V. After scattering from crystal C they are collected by detector D.

Now, you may:

Fix the angle and measure the intensity for different values of the potential

Fix the potential and measure the intensity for different angles



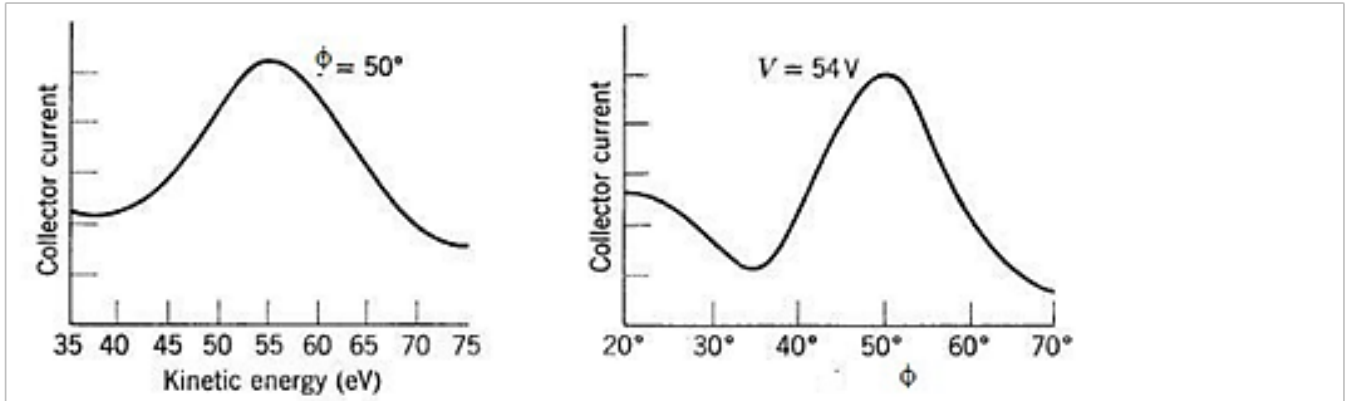
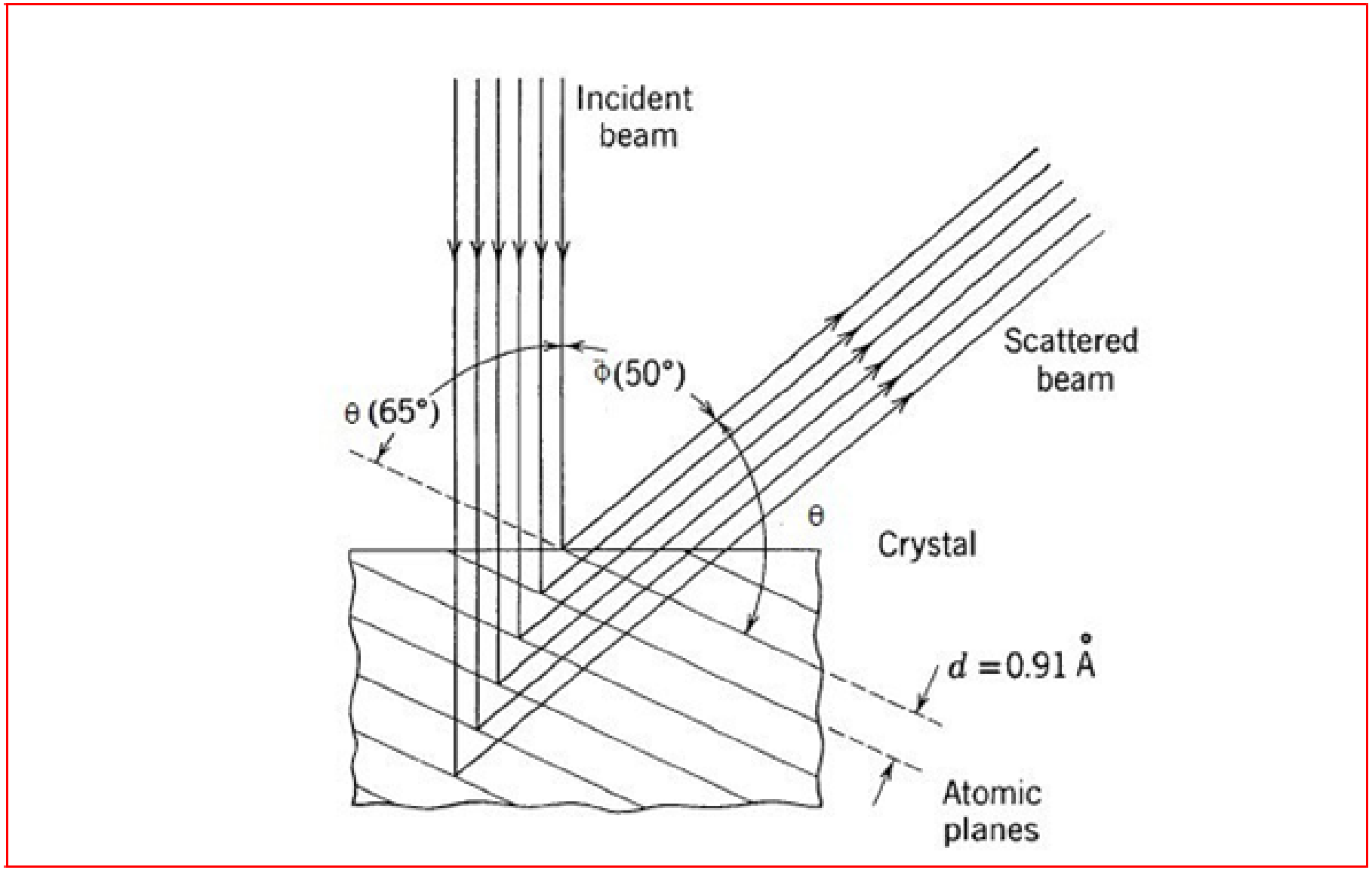


Figure 3-2 *Left:* The collector current in detector *D* of Figure 3-1 as a function of the kinetic energy of the incident electrons, showing a diffraction maximum. The angle ϕ in Figure 3-1 is adjusted to 50°. If an appreciably smaller or larger value is used, the diffraction maximum disappears. *Right:* The current as a function of detector angle for the fixed value of electron kinetic energy 54 eV.

There is a strong scattered electron beam at $\phi = 50°$ for V=54 Volts

It can NOT be understood on the basis of classical particle motion !!

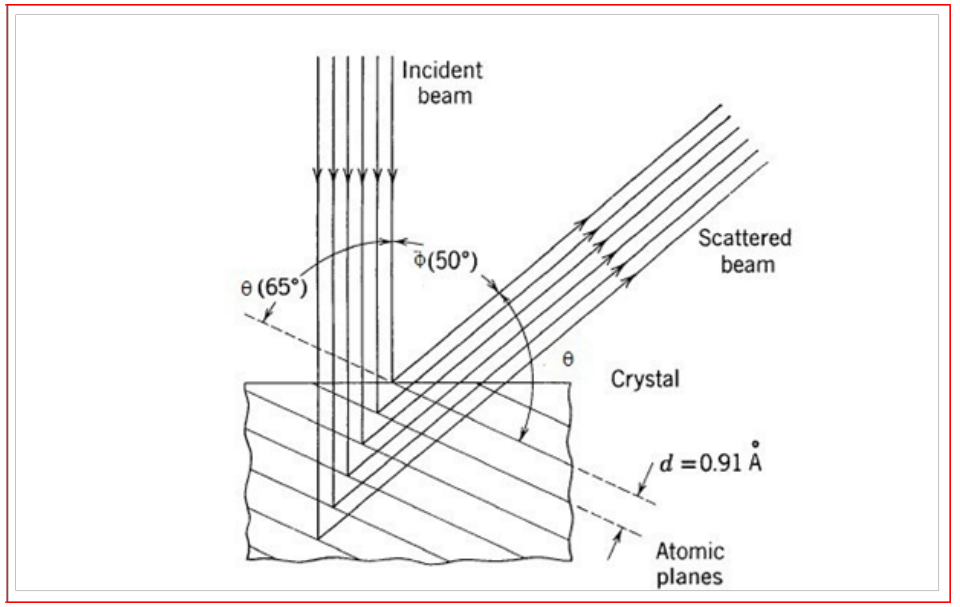
Davisson and Germer's experiment (1927)



Davisson and Germer's experiment. Results

The strong scattering at $\phi = 50°$ for V=54 Volts can only be explained as a constructive interference of waves scattered by the periodic arrangement of the atoms into planes of the crystal.

Same as Bragg reflections of X-rays by the atomic planes of a crystal.



For a Ni crystal d=0.91 Å and $\phi = 50°$

Also $2\theta + \phi = 180°$. Therefore $\theta = 90 - \frac{50}{2} = 65°$.

Recall Bragg's relation. Take n=1

$$\lambda = 2d \sin\theta = 2 \cdot 0.91 \cdot \sin 65 = 1.65 \text{ Å}$$

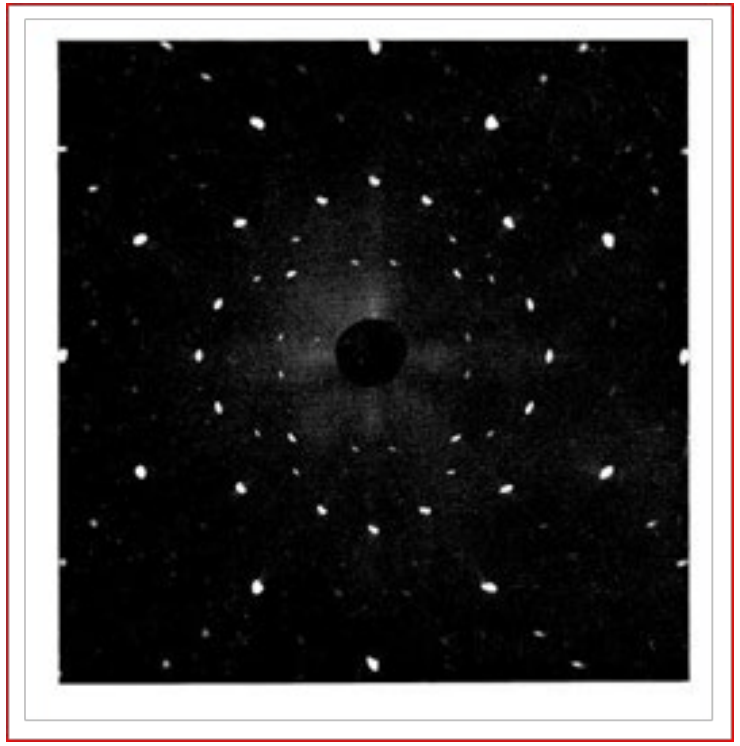
On the other hand, for electrons accelerated by a potential = 54 eV

$$p = \sqrt{2mK} = 3.97\ 10^{-24}\ kg\frac{m}{s} \qquad \lambda = h/p = 1.65\ 10^{-10} m = 1.65 \text{ Å}$$

Therefore, Davisson & Germer's experiment confirms the existence of a wave associated to the motion of particles.

Wave-particle duality in material particles

X-rays



Neutrons

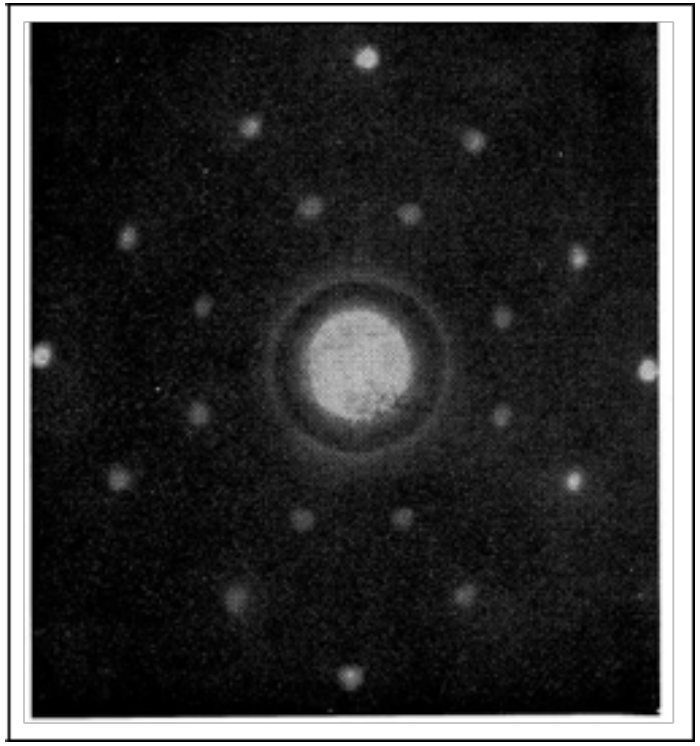


Figure 3-5 *Top:* Laue pattern of x-ray diffraction by a single sodium choride crystal. *Bottom:* Laue pattern of diffraction of neutrons from a nuclear reactor by a single sodium choride crystal.

Wave-particle duality in material particles

- The interference is NOT between waves associated with different electrons.
- The interference is between different parts of the wave associated with a single electron.

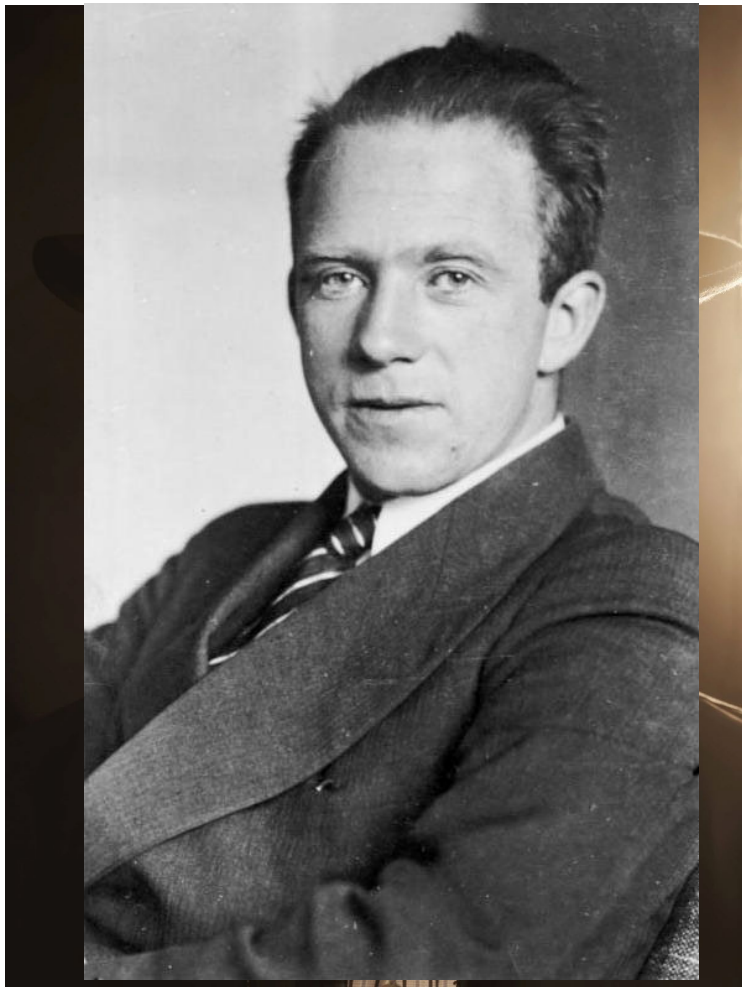
  Demonstrated by sending the electrons one by one.
- **All** material objects show wave-like characteristics. Experiments with beams of H_2, He and slow neutrons
- We need relatively large λ in order to show the wave nature of matter. But in the macroscopic world $p = mv$ is usually a large number because m is large and then λ is very small
- In the microscopic world m is small which implies p small and then λ may be large enough.
- Put the blame on h ! If $h \rightarrow 0$, then $\lambda \rightarrow 0$ and there would be NO waves.

Heisenberg's uncertainty principle

In classical physics, the fundamental laws (Newton's) are deterministic.

Consider a particle

- At t=0 we know its position and momentum (initial conditions)
- Evolution of the particle? Apply Newton's equations
- At a later time, we know exactly the new position and momentum.
- Any doubt? Check it by measurements (these do not perturb the system)
- Is it possible to perform the same process with microscopic particles?



Heisenberg's uncertainty principle

When making observations, there are interactions with the system. But classical physics assumes that the disturbances are very small OR that they can accurately be taken into account.

Suppose we want to measure the distance to the Moon. It can be done by bouncing a radar signal on it. This disturbs the motion of the Moon, but it is negligible.

On a smaller, macroscopic scale, it was assumed that such disturbances were small or at least controllable and they could be taken into account by suitable calculations.

Consequently, it was assumed that the same procedure would apply when dealing with microscopic systems.

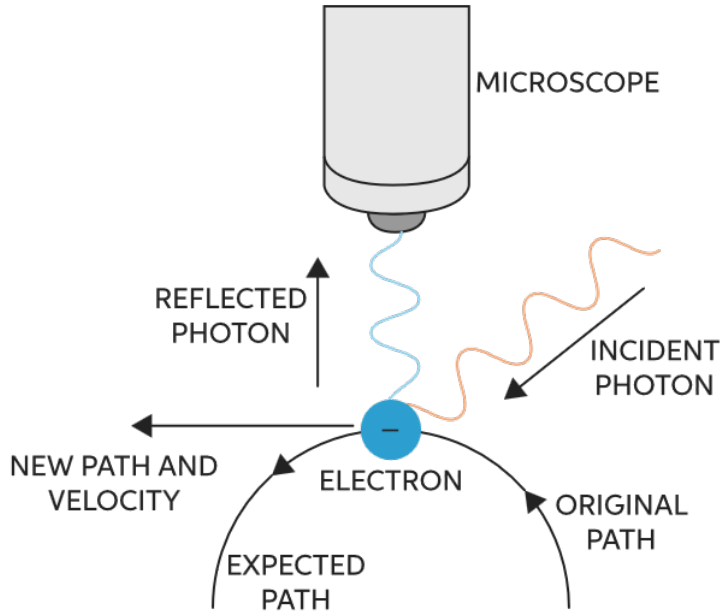
BUT .....

Heisenberg's uncertainty principle

- Is it possible to perform the same process with microscopic particles?

- Is it possible to determine, by an experiment, at the SAME instant both the position and momentum of the particle?

NO !!!!!!!!!!



- **More precisely:**
  Not more accurately than is allowed by Heisenberg's uncertainty (indeterminacy) principle

Heisenberg's uncertainty principle

Can we determine by an experiment, at the SAME instant, both the position and momentum of a particle?

1) Simultaneous measurement of position and momentum

Experiment can not simultaneously determine, in the same experiment, the exact value of a component of momentum p_x and also the exact value of its corresponding coordinate x

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

This also applies to the other components Δp_y and Δy , Δp_z and Δz , so in 3-D we have

$$\Delta x \Delta p_x \geq \frac{\hbar}{2} \qquad \Delta y \Delta p_y \geq \frac{\hbar}{2} \qquad \Delta z \Delta p_z \geq \frac{\hbar}{2}$$

This restriction is NOT on the accuracy to which x or p_x can be determined, but on the product $\Delta x\ \Delta p_x$ in a simultaneous measurement of BOTH.

Heisenberg's uncertainty principle

2) Measurement of the energy *E* and the time *t* required (or available)

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

Δt stands for: (1) the time interval that is needed to observe some change in any physical magnitude of the system or (2) the time interval available for the measurement.

A measurement of the energy with an uncertainty ΔE must take a time interval of at least $\Delta t \geq \hbar/2\Delta E$. In other words, if a system remains in a certain state for a time Δt, then its energy has an uncertainty of at least $\Delta E \geq \hbar/2\Delta t$

This principle has nothing to do with possible improvements in the accuracy of the measurement devices. It would also apply when using IDEAL devices.

Heisenberg's uncertainty principle

Consider a (free) electron moving along the *x*-axis, so $E = \frac{p_x^2}{2m}$

If p_x is known with an indeterminacy Δp_x, then $\Delta \mathrm{E} = \frac{\mathrm{p_x}}{\mathrm{m}} \Delta \mathrm{p_x} = \mathrm{v_x} \Delta \mathrm{p_x}$

We now measure its position, and it takes a time Δt. Then the uncertainty in its position will be $\Delta \mathrm{x} = \mathrm{v_x} \Delta \mathrm{t}$. Therefore

$$\Delta E \Delta t = v_x \Delta p_x \frac{1}{v_x} \Delta x = \Delta p_x \Delta x \geq \frac{\hbar}{2}$$

Consider a particle m=100 g, v=60 m/s, Δv=0.001 m/s

$$\Delta p = m \Delta v \approx 10^{-4} \qquad \Longrightarrow \qquad \Delta x \geq \frac{\hbar}{2\Delta p} \approx 10^{-30}\, m$$

which is negligible at a macroscopic level.

Heisenberg's uncertainty principle

About its physical origin: a thought experiment proposed by Bohr,

We want to measure as accurately as possible the position of a "point" particle, e.g. an electron.

To achieve the greatest precision, we use a microscope to view the electron. The image is sent to a plate.

To see the electron, we must illuminate it

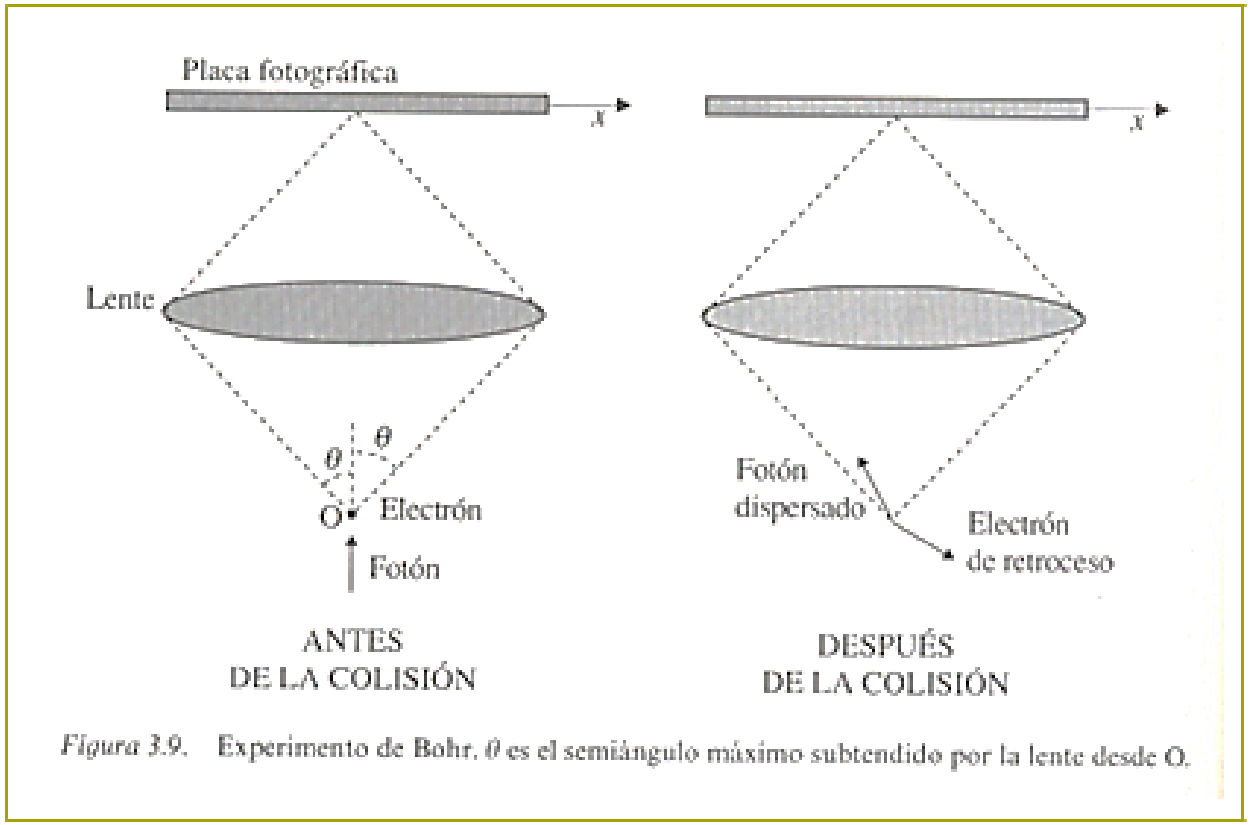
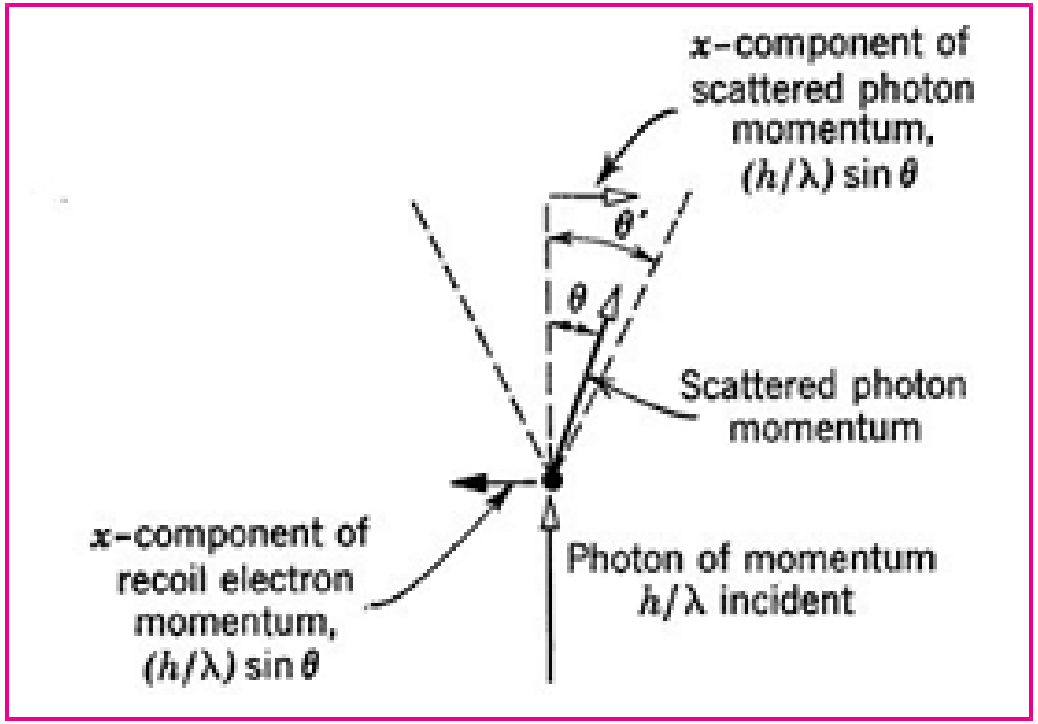


Figura 3.9. Experimento de Bohr. θ es el semiángulo máximo subtendido por la lente desde O.

Heisenberg's uncertainty principle



To see the electron, we must illuminate it. It is the photon scattered by the electron what the observer (plate) sees.

Suppose we can determine its position from **just one** scattered photon.

The scattered photon will be "observed" **IF** it enters the lens. Therefore, the *x*-component of the momentum of the scattered photon must be within the range $p \sin\theta'$ and $-p \sin\theta'$

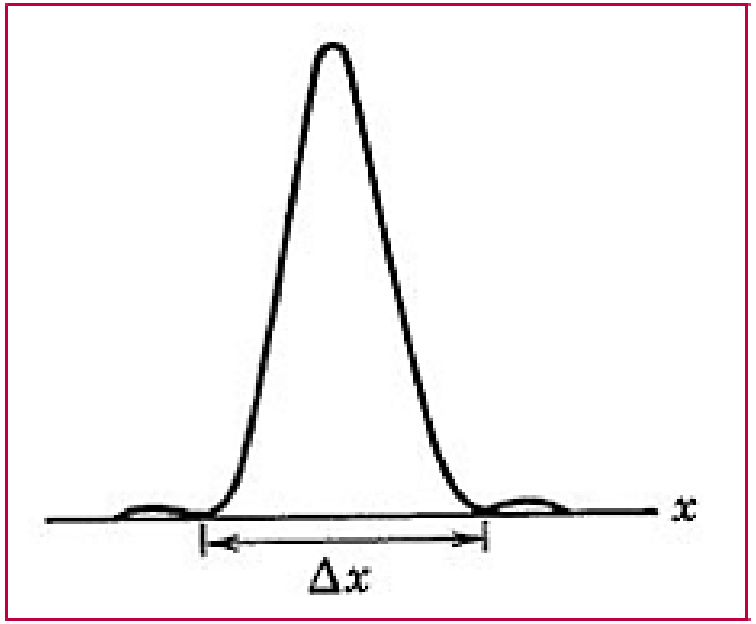
The electron's momentum (its *x*-component) will also be modified, after the collision, by the same amount as the photon's momentum. Therefore, the electron's momentum will have an uncertainty

$$\Delta p_x \sim 2\left(\frac{h}{\lambda}\right)\sin\theta'$$

Heisenberg's uncertainty principle

On the other hand, it is known that a microscope's image of a point is not a point, but a diffraction pattern whose width can be given by

$$\Delta x \sim \frac{\lambda}{\sin\theta'}$$



Therefore

$$\Delta p_x \Delta x \sim \left(\frac{2h\sin\theta'}{\lambda}\right)\left(\frac{\lambda}{\sin\theta'}\right) \sim 2h$$

This relationship has a universal character. It says that it is impossible, **in the SAME experiment**, to determine **simultaneously** the position and momentum of a particle with total precision. The uncertainty in both measurements is related:

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

Heisenberg's uncertainty principle

The electron's momentum after the collision may take any value compatible with the conservation laws.

We may reduce Δp_x by increasing λ or decreasing θ'. But this procedure increases Δx .

The problem stems from the quantization of radiation. We need at least a photon ($p = h/\lambda$). Indeed, this is the problem!

In classical physics it was possible to use a momentum as small as we wanted and at the same time to use a very small λ in order to improve the resolution.

NOT ANYMORE !!!!!

Heisenberg's uncertainty principle

The previous argument can be also made for Δp_y y Δy , Δp_z y Δz , therefore in 3-D we have

$$\Delta x \Delta p_x \geq \frac{\hbar}{2} \qquad \Delta y \Delta p_y \geq \frac{\hbar}{2} \qquad \Delta z \Delta p_z \geq \frac{\hbar}{2}$$

This principle, Heisenberg's uncertainty principle, is closely connected to the fact that it is impossible to measure something without disturbing it.

Moreover, it also connects other pairs of measurable magnitudes: energy-time, x and y components of the angular moment, etc

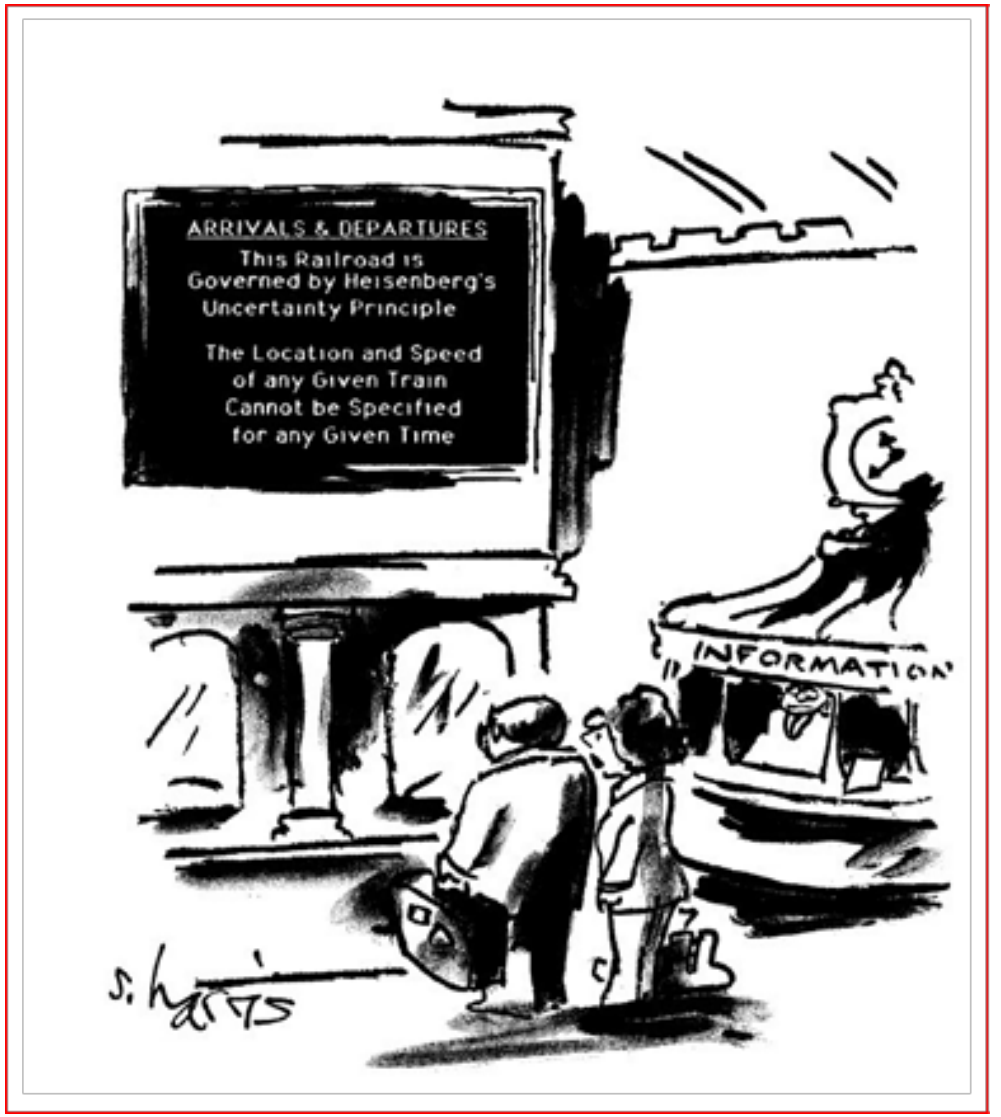
$$\Delta E \Delta t \geq \frac{\hbar}{2}, \ldots$$

But you can measure a magnitude with as much precision as you like

Heisenberg's uncertainty principle

Consequences:

- We must abandon the idea of a continuous trajectory
- We can not determine **simultaneously** the position and momentum of a subatomic particle at a given time *t*; therefore, it is impossible to predict with total accuracy the behavior of the particle for all future time.
- The classical determinism does no longer exist. We can only make predictions about the probable behavior of the particle.
- When there are several identical particles, i.e., electrons, it will be **impossible** to distinguish among them.



Matter waves

Particle moving along the *x*-axis. No force acting on it.

de Broglie: There is a matter wave associated to a particle. What does it look like?

Suppose that at *t*=0, we can measure the amplitude of the matter wave.

The amplitude should be non-zero in a region around the particle.

It should move along the *x*-axis with the same velocity as the particle

de Broglie assumes the same equations for photons and for particles

$$E = h\nu \qquad\qquad p = h/\lambda$$

which, written in terms of the angular frequency $\omega = 2\pi\nu$ and the wavenumber $k = 2\pi/\lambda$ take the form

$$E = \hbar\omega \qquad\qquad p = \hbar k$$

But we have different *E* expressions for radiation and for particles

Radiation $E = pc \implies \omega = kc$ Particles (Non-Relativ) $E = \frac{p^2}{2m} \implies \omega = \frac{\hbar k^2}{2m}$

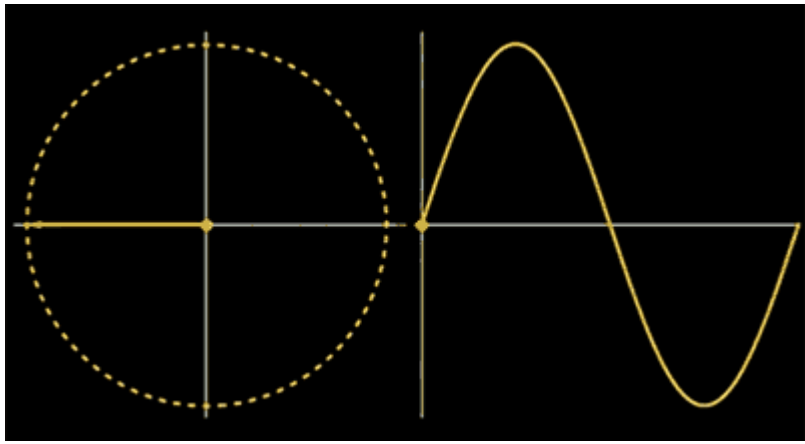
Matter waves

Simplest type of wave motion. Sinusoidal wave

Constant amplitude (unity) moving with uniform velocity in the direction of increasing x-values

$$\Psi(x,t) = \sin(kx - \omega t) \qquad k = \frac{2\pi}{\lambda} \qquad \omega = 2\pi\nu$$

$$\Psi(x,t) = \sin\left[2\pi\left(\frac{x}{\lambda} - \nu t\right)\right] \qquad \Psi(x,t) = \sin[2\pi(\kappa x - \nu t)], \quad \kappa = \frac{1}{\lambda}$$



Interpretation:

Fix x**:** Ψ oscillates in time sinusoidally with **frequency** ν and amplitude 1.

Fix t**:** Ψ varies sinusoidally with **wavelength** λ (or reciprocal wavelength / wavenumber κ).

Zeros (nodes): at positions $x_n$satisfying

$$2\pi(\kappa x_n - \nu t) = n\pi \qquad (n = 0, \pm 1, \pm 2, \dots) \qquad x_n = \frac{n}{2\kappa} + \frac{\nu}{\kappa}t \qquad w = \frac{dx_n}{dt} = \frac{\nu}{\kappa} = \lambda\nu$$

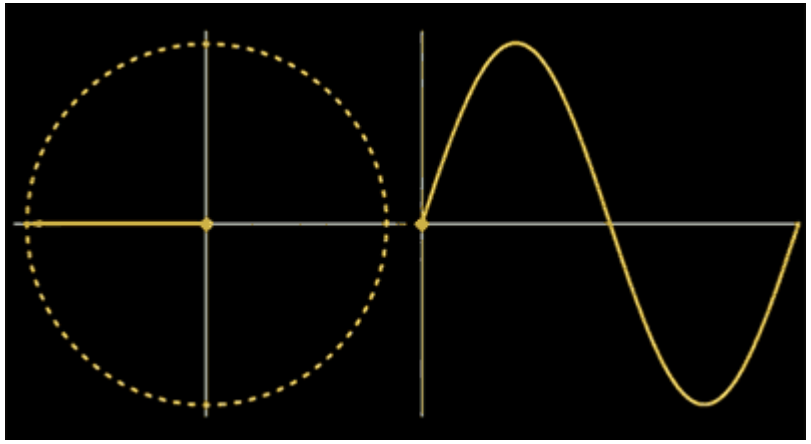
Matter waves

Simplest type of wave motion. Sinusoidal wave

Constant amplitude (unity) moving with uniform velocity in the direction of increasing x-values

$$\Psi(x,t) = \sin(kx - \omega t)$$

$$k = \frac{2\pi}{\lambda} \qquad \omega = 2\pi\nu$$



Wave velocity, w $\quad w = \frac{dx_n}{dt} = \frac{\nu}{\kappa}$

$$\left(\kappa = \frac{1}{\lambda}\right) \qquad w = \lambda\nu$$

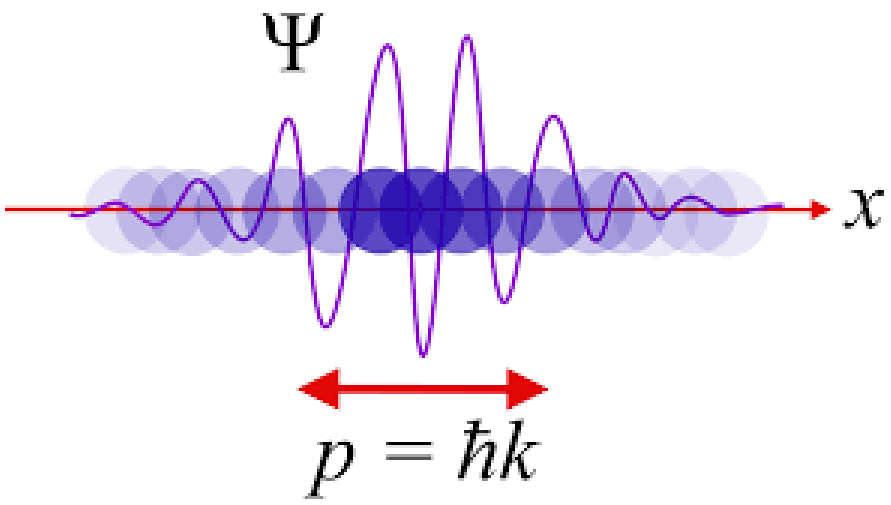
$$(E = h\nu, p = h/\lambda) \qquad w = \frac{h}{p}\frac{E}{h} = \frac{E}{p}$$

$$\left(E = \frac{p^2}{2m}, p = mv\right) \quad w = \frac{v}{2}$$

Matter wave cannot keep up with the particle it controls?!

Matter waves

Intensity of wave ∝ Probability of finding particle



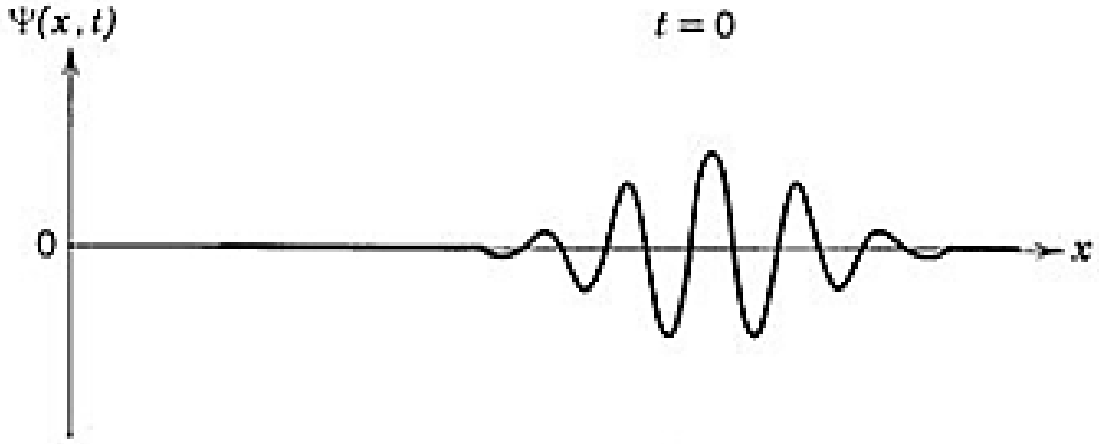


Figure 3-7 A de Broglie wave for a particle.

Matter waves

Group of waves. Add an infinitely large number of sinusoidal waves.

Let's start by adding two waves (with infinitesimally differing frequencies ν and inverse wavelengths κ)

$$\Psi(x,t) = \Psi_1(x,t) + \Psi_2(x,t)$$

$$\Psi_1(x,t) = \sin 2\pi(\kappa x - \nu t) \qquad \Psi_2(x,t) = \sin 2\pi[(\kappa + d\kappa)x - (\nu + d\nu)t]$$

But $$\sin A + \sin B = 2\cos[(A-B)/2]\sin[(A+B)/2]$$

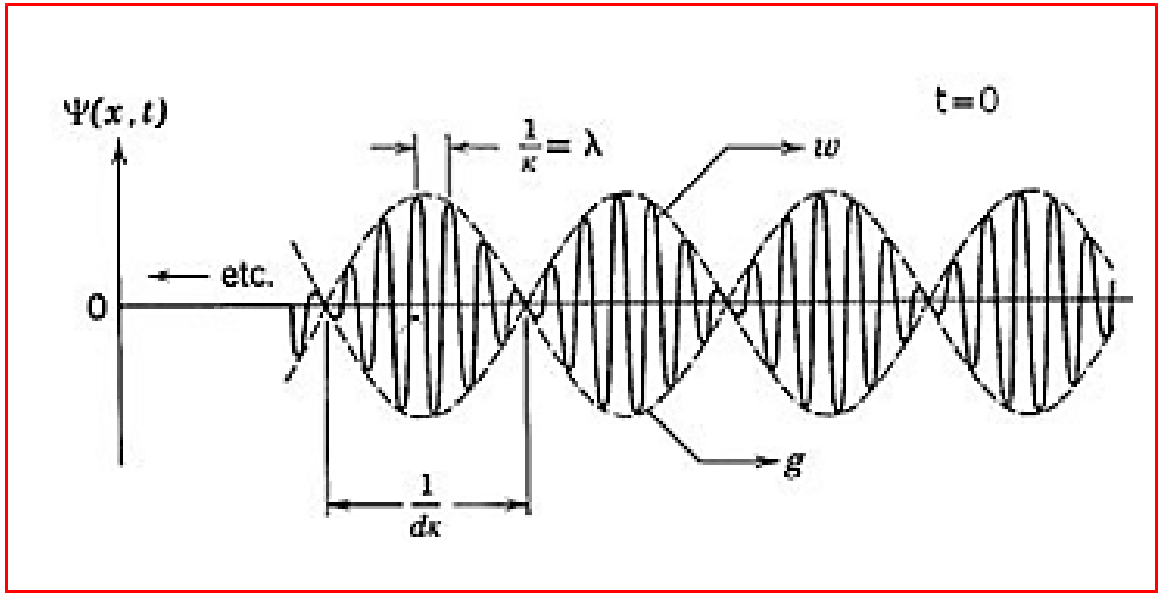
So, we have $$\Psi(x,t) = 2\cos 2\pi\left[\frac{d\kappa}{2}x - \frac{d\nu}{2}t\right]\sin 2\pi\left[\frac{2\kappa + d\kappa}{2}x - \frac{2\nu + d\nu}{2}t\right]$$

Matter waves

Taking into account that

$$d\nu \ll 2\nu \qquad\qquad d\kappa \ll 2\kappa$$

$$\Psi(x,t) = 2\cos 2\pi\left[\frac{d\kappa}{2}x - \frac{d\nu}{2}t\right]\sin 2\pi(\kappa x - \nu t)$$



The second term in Ψ(*x*, *t*) is a sinusoidal wave

The first term modulates the amplitude of the oscillations.

Group velocity $$g = \frac{d\nu/2}{d\kappa/2} = \frac{d\nu}{d\kappa} = \frac{d\omega}{dk}$$

Matter waves

Combine an infinite number of sinusoidal waves to form a group. The **wave velocity** and **group velocity** have the same expressions as before.

$$w = \frac{\nu}{\kappa} \qquad g = \frac{d\nu}{d\kappa} = \frac{d\omega}{dk}$$

Free particle: Evaluate the group velocity of the matter waves associated with the particle

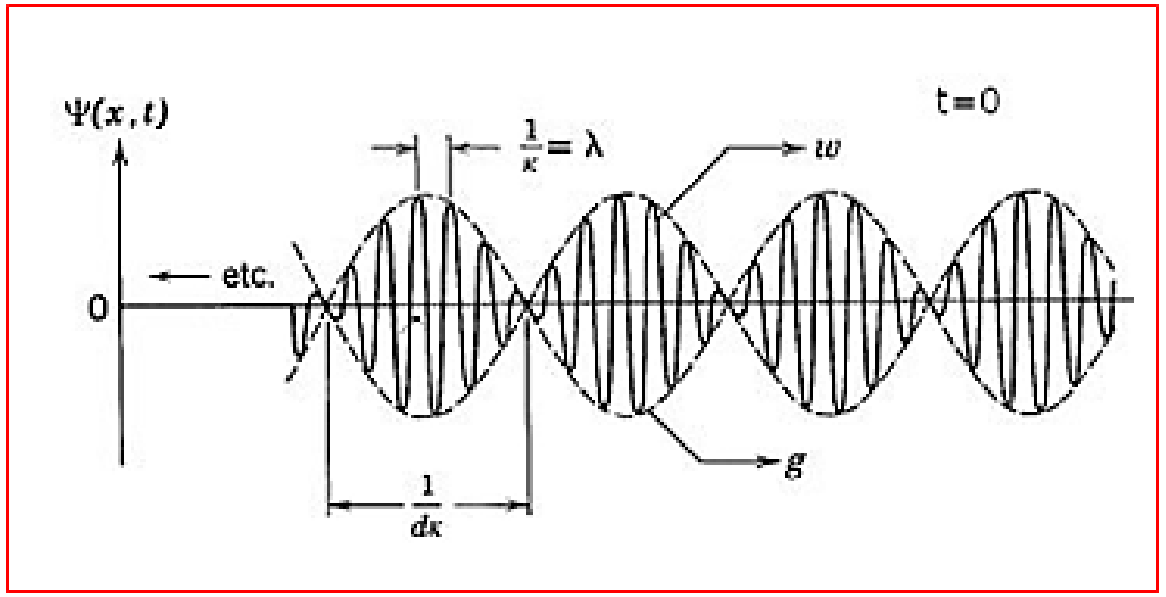
According to de Broglie $E = h\nu$ $p = \frac{h}{\lambda} = h\kappa$ $dE = hd\nu$ $dp = hd\kappa$

Group velocity $g = \frac{d\nu}{d\kappa} = \frac{dE}{dp}$

Particle $\mathrm{E} = \frac{\mathrm{mv}^2}{2}$ $\mathrm{p} = \mathrm{mv}$ $\frac{\mathrm{dE}}{\mathrm{dp}} = \frac{\mathrm{mvdv}}{\mathrm{mdv}} = \mathrm{v} = \mathrm{g}$

The velocity of the **group** of matter waves is equal to the velocity of the particle

Matter waves



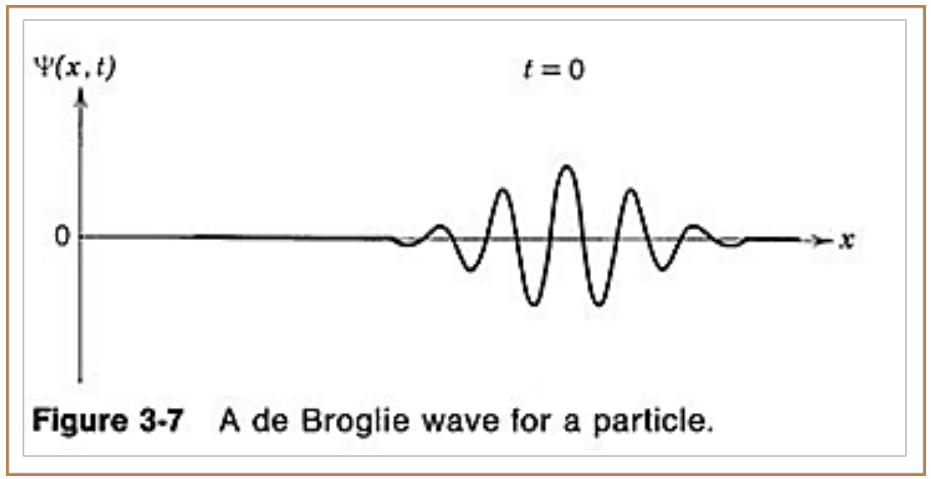
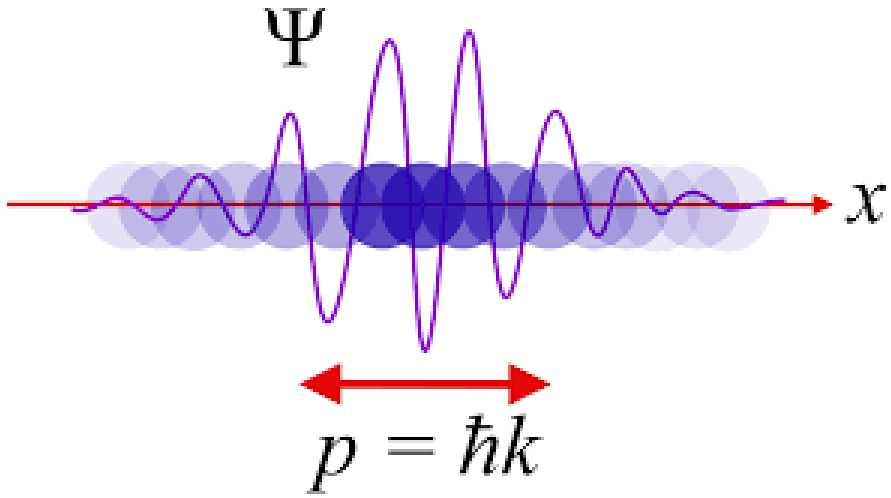


Figure 3-7 A de Broglie wave for a particle.

Intensity of wave ∝ Probability of finding particle



Matter waves

Combine a greater number of sinusoidal waves to form a group.

$$\Psi(x,t) = \sum_k A_k \cos 2\pi(kx - \nu t)$$

slightly different wavelengths $\kappa = 1/\lambda$ $\kappa = 9, 10 \ldots 15$

Amplitudes A_κ

$A_{12} = 1$, $A_{13} = A_{11} = 1/2$, $A_{14} = A_{10} = 1/3$, $A_{15} = A_9 = 1/4$

Analyze the superposition at t=0

- The waves are in phase at $x = 0$
- They get out of phase with increasing (and decreasing) x
- We observe a central group and others at greater distances than before
- Superposing an infinite number of waves gives one group **only**.

